Supply Chain Resilience Technology Adoption Analysis

Report generated on 2025-05-13

Introduction

This report presents findings from a survey assessing the impact of emerging technologies on supply chain resilience. The analysis covers technology adoption patterns, effectiveness in mitigating disruptions, barriers to implementation, and the role of collaboration in enhancing supply chain resilience.

Methodology

The analysis included both quantitative and qualitative methods:
- Hypothesis testing (t-tests, chi-square, ANOVA, correlation)
- Descriptive statistics and visualizations
- Qualitative analysis of open-ended responses
- Examination of relationships between technology adoption and resilience metrics

Key Findings

Data Loading

Successfully loaded data with 45 records and 68 columns.

Initial Data Exploration

{'Missing Values': 1023, 'Duplicate Records': 0, 'Data Types': {dtype('O'): 37, dtype('bool'): 15, dtype('float64'): 11, dtype('int64'): 5}, 'Numeric Columns': ['response_id', 'last_page', 'seed', 'to_what_extent_has_your_company_implemented_the_following_technologies_likert_scale_1_not_implemented_5_fully_implemented_ai', 'to_what_extent_has_your_company_implemented_the_following_technologies_likert_scale_1_not_implemented_5_fully_implemented_iot', 'to_what_extent_has_your_company_implemented_the_following_technologies_likert_scale_1_not_implemented_5_fully_implemented_blockchain', 'to_what_extent_has_your_company_implemented_the_following_technologies_likert_scale_1_not_implemented_5_fully_implemented_cloud_computing', 'to_what_extent_has_your_company_implemented_the_following_technologies_likert_scale_1_not_implemented_5_fully_implemented_robotics_amp_automation', 'to_what_extent_has_your_company_implemented_the_following_technologies_likert_scale_1_not_implemented_5_fully_implemented_big_data_analytics', 'what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_other', 'to_what_extent_have_new_technologies_improved_communication_and_coordination_with_your_suppliers_and_partn

ers_1_not_at_all_2_slightly_3_moderately_4_highly_5_very_high',
'how_effectively_have_new_technologies_helped_in_information_sharing_and_real_time_decision_making_across_your_supply_chain_1_not_at_all_2_slightly_3_moderately_4_highly_5_very_high', 'disruption_count', 'tech_adoption_count', 'resilience_score', 'company_size_num'], 'Categorical Columns': ['date_submitted', 'start_language', 'what_industry_does_your_company_belong_to', 'what_industry_does_your_company_belong_to_other', 'what_is_your_role_in_the_company', 'what_is_your_role_in_the_company_other', 'what_is_the_size_of_your_company', 'natural_disasters', 'supplier_failures', 'cybersecurity_threats', 'global_crises', 'transportation_delays', 'which_types_of_disruptions_have_significantly_impacted_your_supply_chain_in_the_past_five_years_select_all_that_apply_other',
'how_did_your_company_respond_to_these_disruptions_select_all_that_apply_increased_inventory_buffers',
'how_did_your_company_respond_to_these_disruptions_select_all_that_apply_diversified_suppliers',
'how_did_your_company_respond_to_these_disruptions_select_all_that_apply_implemented_new_technologies',
'how_did_your_company_respond_to_these_disruptions_select_all_that_apply_other',
'which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_artificial_intelligence_ai',
'which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_internet_of_things_iot',
'which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_blockchain',
'which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_cloud_computing',
'which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_robotics_amp_automation',
'which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_big_data_analytics',
'which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_simulation', 'which_technologies_does_your_company_currently_use_for_supply_chain_resilience_select_all_that_apply_other',
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'which_of_these_technologies_have_had_the_most_positive_impact_on_supply_chain_resilience_rank_from_most_to_least_effective_rank_3',
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'which_of_these_technologies_have_had_the_most_positive_impact_on_supply_chain_resilience_rank_from_most_to_least_effective_rank_6',
'which_of_these_technologies_have_had_the_most_positive_impact_on_supply_chain_resilience_rank_from_most_to_least_effective_rank_7',
'if_you_selected_other_in_the_previous_question_please_specify_the_technology_otherwise_you_may_leave_this_blank',
'what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_high_implementation_costs',
'what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_lack_of_skilled_workforce', 'what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_integration_issues',
'what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_resistance_to_change',
'which_technologies_have_enhanced_collaboration_in_your_supply_chain_cloud_based_platforms_e_g_erp_scm_software', 'which_technologies_have_enhanced_collaboration_in_your_supply_chain_blockchain_for_transparency',

'which_technologies_have_enhanced_collaboration_in_your_supply_chain_ai_powered_predictive_analytics', 'which_technologies_have_enhanced_collaboration_in_your_supply_chain_iot_for_real_time_tracking', 'which_technologies_have_enhanced_collaboration_in_your_supply_chain_digital_twin_amp_simulations', 'which_technologies_have_enhanced_collaboration_in_your_supply_chain_other', 'what_are_the_biggest_barriers_preventing_further_adoption_of_these_technologies_select_all_that_apply_financial_constraints', 'what_are_the_biggest_barriers_preventing_further_adoption_of_these_technologies_select_all_that_apply_technical_challenges', 'what_are_the_biggest_barriers_preventing_further_adoption_of_these_technologies_select_all_that_apply_regulatory_issues', 'what_are_the_biggest_barriers_preventing_further_adoption_of_these_technologies_select_all_that_apply_uncertainty_about_roi', 'what_are_the_biggest_barriers_preventing_further_adoption_of_these_technologies_select_all_that_apply_other', 'are_there_any_emerging_technologies_your_company_is_considering_for_future_adoption', 'do_you_have_any_further_insights_on_how_technology_can_improve_supply_chain_resilience', 'industry_group', 'high_tech_company']}

Data Cleaning

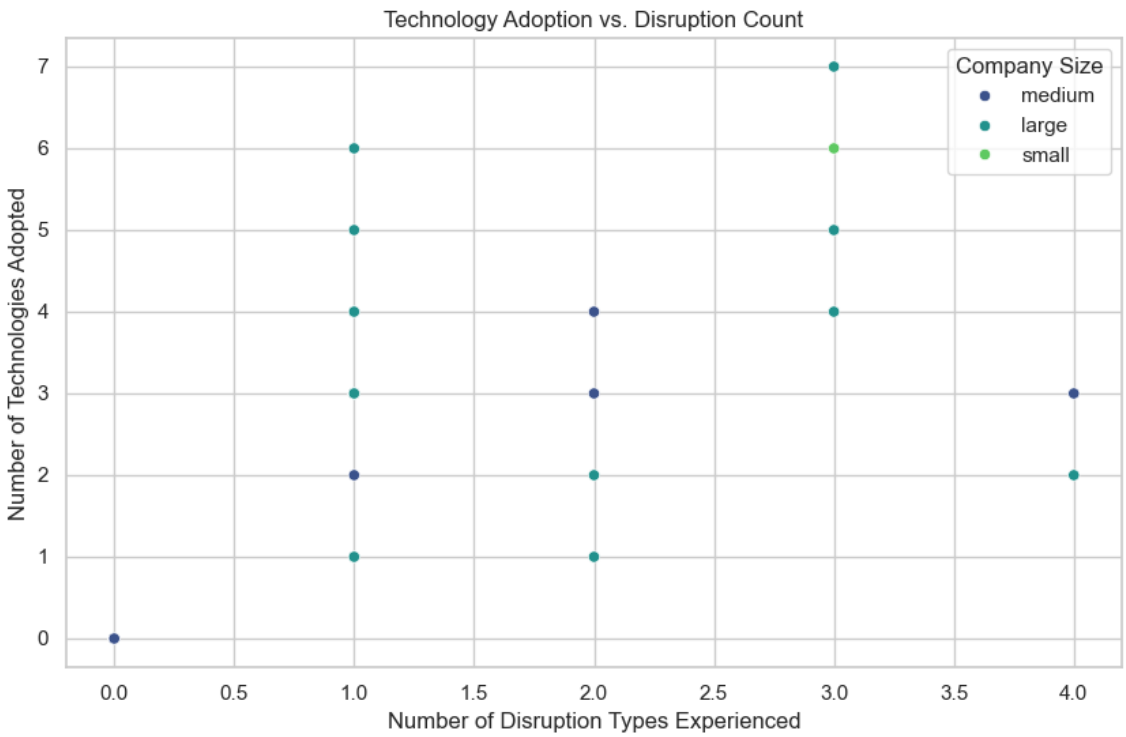
Completed data cleaning including handling missing values, standardizing categories, and creating derived metrics.

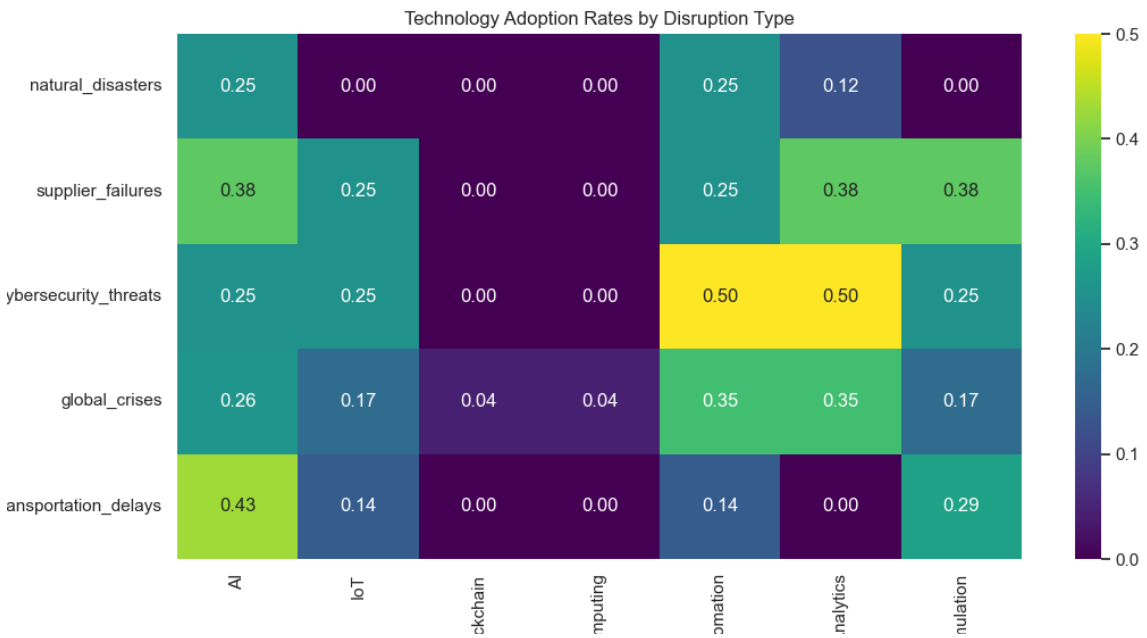
Data Distribution Visualization

Created visualizations showing industry distribution, company size, disruption counts, and technology adoption.

H1: Disruptions and Supply Chain Resilience

Hypothesis: H1a: Higher disruptions lead to more technology adoption
Correlation: 0.6833043547752813
P-value: 2.3117849130403122e-07
Sample Size: 45
Conclusion: There is a significant relationship between disruption count and technology adoption (r=0.68, p=0.0000).





H1: Disruptions and Supply Chain Resilience - H1b

No significant associations found between specific disruption types and technology choices.

H2: Adoption of Technologies - H2a

Technology: AI
Conclusion: Insufficient data for chi-square test

Technology: IoT
Conclusion: Insufficient data for chi-square test

Technology: Big Data Analytics
Conclusion: Insufficient data for chi-square test

H2: Adoption of Technologies - H2b

Technology: Cloud Computing
Conclusion: Insufficient data for chi-square test

Technology: Blockchain
Conclusion: Insufficient data for chi-square test

H3: Effectiveness of Technologies - H3a

Technology: AI
Mean Resilience Score - Users: 3.2142857142857144
Mean Resilience Score - Non-users: 1.6578947368421053
T-statistic: 2.3208564868456665
P-value: 0.04400006168020398
Conclusion: Companies using AI have significantly different resilience scores

Technology: IoT
Mean Resilience Score - Users: 3.625
Mean Resilience Score - Non-users: 1.7317073170731707
T-statistic: 3.6325563405769214
P-value: 0.009402031291950441
Conclusion: Companies using IoT have significantly different resilience scores

H3: Effectiveness of Technologies - H3b

Conclusion: Insufficient data to perform ANOVA (need at least two groups with sufficient data)

H4: Barriers to Adoption - H4a

Barrier Prevalence: {'what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_high_implementation_costs': 10, 'what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_lack_of_skilled_workforce': 10, 'what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_integration_issues': 9, 'what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_resistance_to_change': 5}

Primary Barrier: what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_high_implementation_costs
Conclusion: The most common barrier is what_were_the_main_challenges_in_implementing_these_technologies_select_all_that_apply_high_implementation_costs

H4: Barriers to Adoption - H4b

Technology: AI
Conclusion: Insufficient data for chi-square test

Technology: Big Data Analytics
Conclusion: Insufficient data for chi-square test

H5: Collaboration and Stakeholder Engagement - H5a

Correlation: 0.8735079610336864
P-value: 1.1116605966604523e-07
Sample Size: 22
Conclusion: There is a significant relationship between collaboration and resilience (r=0.87, p=0.0000).

H5: Collaboration and Stakeholder Engagement - H5b

Correlation: -0.24378260305040292
P-value: 0.2742636902905429
Sample Size: 22
Conclusion: There is no significant relationship between technology adoption and collaboration (r=-0.24, p=0.2743).

H6: Control Mechanisms and Decision-Making - H6a

Mean Resilience - AI/Analytics Users: 3.4615384615384617
Mean Resilience - Non-users: 1.265625
T-statistic: 4.808492559302943
P-value: 3.091175825650368e-05
Conclusion: Companies using AI/Analytics have significantly different resilience scores

H6: Control Mechanisms and Decision-Making - H6b

Mean Resilience - IoT/Blockchain Users: 3.5
Mean Resilience - Non-users: 1.7
T-statistic: 3.8561970849056197
P-value: 0.00246414434928595
Conclusion: Companies using IoT/Blockchain have significantly different resilience scores

Qualitative Analysis

do_you_have_any_further_insights_on_how_technology_can_improve_supply_chain_resilience: {'sample_responses': ['leider nicht.', 'make machines as work in more than one power.', 'the fast decision making using new technology and ai can solve most of the resiliences.', 'yes', 'i'], 'total_responses': 18, 'wordcloud_generated': True}
if_you_selected_other_in_the_previous_question_please_specify_the_technology_otherwise_you_may_leave_this_blank: {'sample_responses': ['3d druck / additive fertigung', '*', '.', 'big data analysis', 'robitics'], 'total_responses': 14, 'wordcloud_generated': True}
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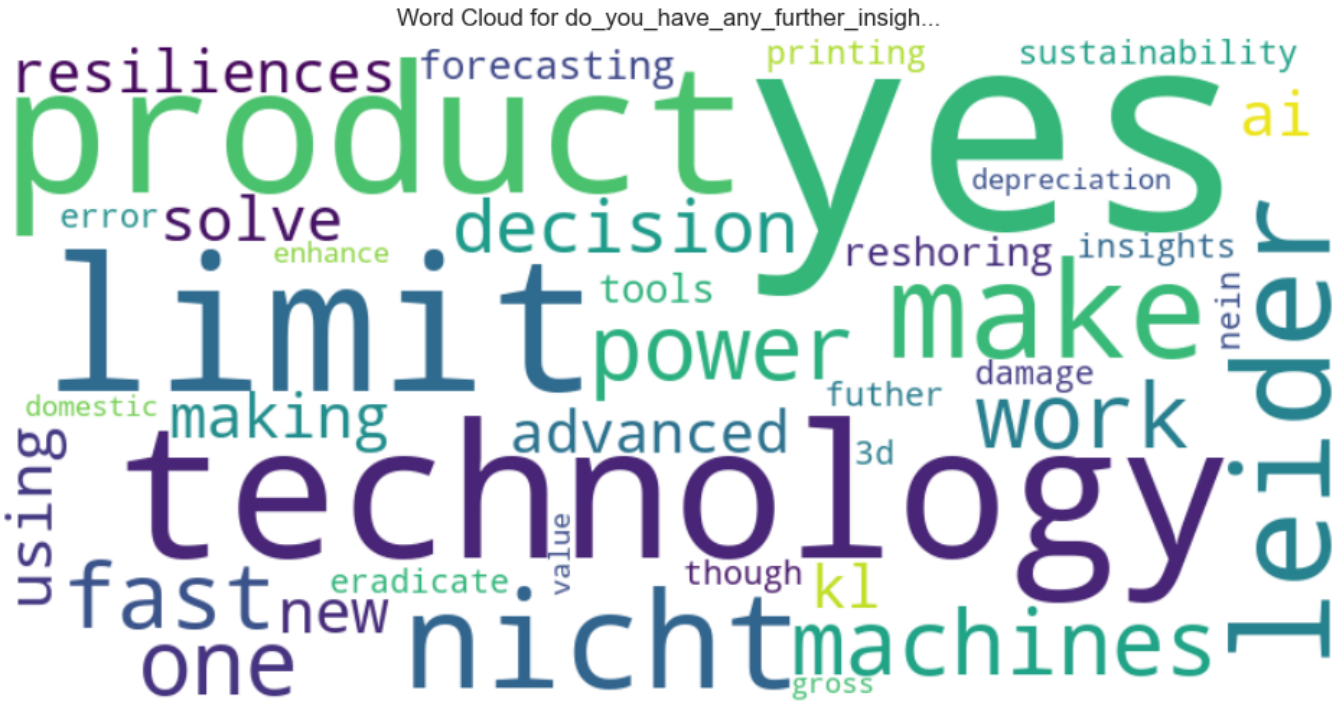
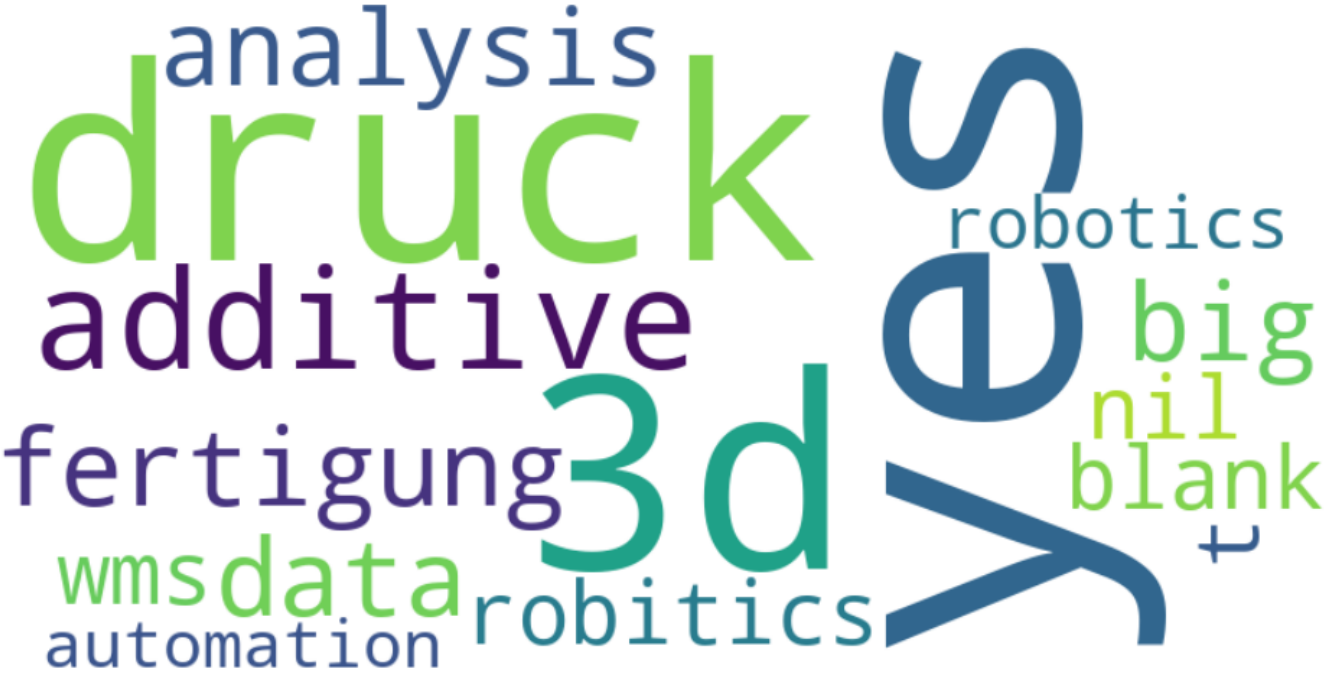
what_are_the_biggest_barriers_preventing_further_adoption_of_these_technologies_select_all_that_apply_other: {'sample_responses': ['Überzeugung des Managements', '2', '2', '2', '2'], 'total_responses': 21, 'wordcloud_generated': True}

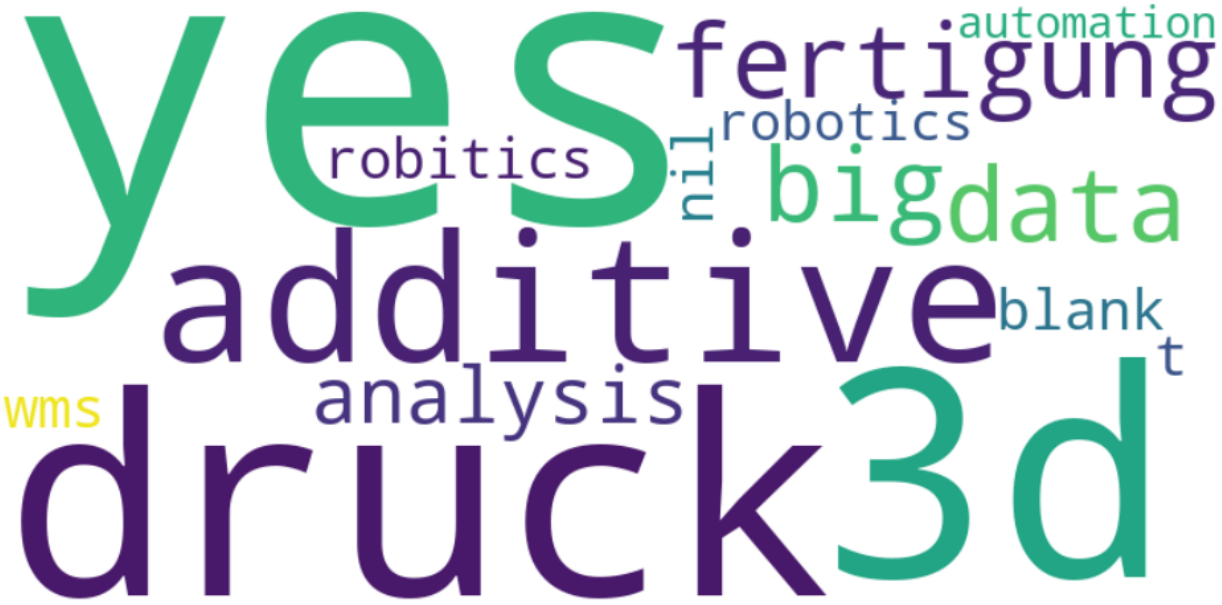


Word Cloud for do_you_have_any_further_insights_on_how_technology_can_improve_supply_chain_resilience

Word Cloud for if_you_selected_other_in_the_p...



Word Cloud for if_you_selected_other_in_the_previous_question_please_specify_the_technology_otherwise_you_may_leave_this_blank



Word Cloud for what_are_the_biggest_barriers_...

Conclusions and Recommendations

Key Recommendations:
1. Prioritize AI and IoT adoption for companies facing frequent disruptions
2. Invest in workforce training to overcome skill barriers
3. Implement technology combinations (AI + Blockchain + Cloud) for maximum resilience
4. Focus on collaboration technologies to enhance supply chain visibility
5. Address cost barriers through phased implementation plans

Future Research Directions:
- Longitudinal study on technology adoption impact
- Industry-specific technology effectiveness
- Ethical implications of AI in supply chain decisions